



Profile Summary

- M.Tech CSE (AI) student at IIIT Delhi with hands-on experience in GenAI systems, RAG pipelines, multi-agent workflows, backend APIs, and scalable AI software development.
- Experienced in building agentic AI systems using MCP-style tool servers, A2A agent communication, orchestrator workflows, evaluation metrics, observability, and production-ready ML pipelines.

Education

Indraprastha Institute of Information Technology, Delhi (IIITD) Masters of Technology CSE– CGPA : 7.6	(2024-Jun 2026) Delhi
Sanskar College of Engineering and Technology, Ghaziabad Dr.A.P.J. Abdul Kalam Technical University Lucknow (AKTU) (UPTU) Bachelors of Technology CSE– CGPA : 8.04	(2016-2020) Ghaziabad, UP

Technical Skills

- **Programming & CS Fundamentals:** Python, C++, Data Structures & Algorithms, Operating Systems (OS), Database Management Systems (DBMS), Computer Networks (CN), Debugging, Reinforcement Learning, Optimization, Simulation
- **Backend & Systems:** FastAPI, REST APIs, Linux, System Debugging, Performance Optimization (basic),MCP-style Tool Servers, A2A Agent Communication, Agent Orchestration, Tool-calling, Trace-level Observability
- **Frameworks & Tools:** Docker, Git, Software Testing, Test Automation, OR-Tools (Learning), Stable-Baselines3 (Learning), MLflow (Learning), NetworkX, SQLite, Docker, Git, FastAPI, HTTPX
- **Databases:** MongoDB, FAISS, SQLite, Vector Databases
- **Machine Learning/ GenAI:** PyTorch, Transformers, Model Optimization, Reinforcement Learning (Learning), Experiment Design, Evaluation Frameworks, RAG, LLM Evaluation, Prompt Engineering, Context Management, Function Call Graphs, Agentic Workflows
- **Full Stack AI/MLOps:** Streamlit (Frontend), React (Learning), End-to-End ML Pipeline Development

Relevant Coursework

• Graduate Algorithms	• Deep Learning	• Operating System	• Machine Learning
• Object Oriented Programming	• Artificial Intelligence	• Computer Networking	• Database Management

Experience

Research Intern – Microbiomes Informatics Lab, IIIT Delhi	(Jan 2025 – June 2026)
• Engineered a Transformer-based protein language model for Genotoxin classification, achieving 98% accuracy and 0.9993 F1-score. • Built scalable data pipelines for protein sequence ingestion, preprocessing, training, and evaluation across large-scale datasets. • Applied contrastive learning, MLM pretraining, and LoRA fine-tuning to improve model performance on ~200B protein sequences. • Performed debugging, error analysis, metric-based validation, and feedback-driven improvements to enhance reliability and reproducibility.	

Projects

Agentic Code Intelligence Platform using MCP and A2A	FastAPI, MCP, A2A Agents, AST, NetworkX, SQLite, Docker	(Apr 2026 – May 2026)
• Built an MCP + A2A based agentic code intelligence for large repository understanding, generating file-level, function-level, dependency-level summaries, API-flow tracing, and function call graph analysis using orchestrator-driven multi-agent workflows with pluggable LLM support. • Designed secure MCP-style tooling for AST parsing, dependency extraction, function analysis, Git diff tracking, SQLite caching, observability, workspace sandboxing, and secret-pattern redaction for scalable and production-oriented repository intelligence pipelines.		

YouTube RAG Chatbot for Video Transcript	LangChain, OpenAI APIs, FAISS, Python	(Jan 2026 – Feb 2026)
• Built an AI-driven retrieval and decision-support pipeline leveraging semantic retrieval, dynamic prompt orchestration, and top-K ranking strategies for context-aware response generation. • Developed a top-K retrieval system with dynamic prompt generation and OpenAI ChatModel integration to generate context-aware, coherent, and semantically relevant responses.		

Prompt-to-Prompt Image Editing with Masked Attention Control	PyTorch, Diffusers, Stable Diffusion	(Aug 2025 – Dec 2025)
• Implemented Prompt-to-Prompt image editing by manipulating cross-attention maps in Stable Diffusion to enable controlled, structure-preserving semantic edits. • Built an attention interception and storage pipeline within UNet to reuse and replace token-level attention across prompts, maintaining spatial consistency. Integrated LocalBlend-based masked attention control to restrict edits to target regions, improving precision and preventing unintended global changes.		

Certifications, Achievements & Responsibility

- **GATE Qualified** — Qualified GATE 2024 in Computer Science and GATE 2025 in Data Science and Artificial Intelligence.
- **Udemy Agentic AI & Generative AI Certification** — Completed *AI A-Z [2026]: Agentic AI, Gen AI, Prompt Engineering and RL* from Udemy, May 2026.
- **Reinforcement Learning Hackathon** — Participated in AMD AI Reinforcement Learning Hackathon by AMD, Yardi School of AI, IIT Delhi, and Unsloth.
- **Kaggle Competitions** — Participated in 3 Kaggle challenge competitions involving AI/ML problem-solving and model development.

Interests

- Decision Intelligence, Optimization Systems, Reinforcement Learning, Generative AI, Scalable AI Systems, MLOps.